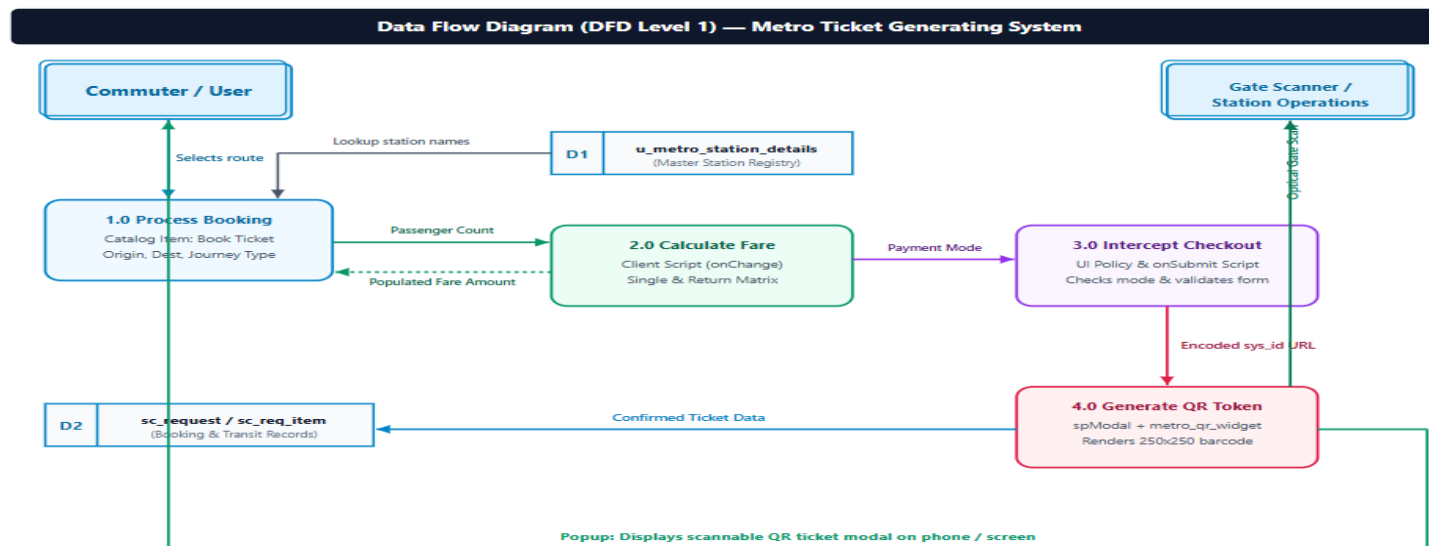


Project Design Phase-II Data Flow Diagram & User Stories

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

1. Commuter navigates to the Service Portal (/sp) and opens the "Book A Metro Ticket" catalog item.
2. Form reads the station master list from the database table (u_metro_station_details).
3. Commuter inputs origin, destination, journey type, and passenger count; an onChange client script calculates fares dynamically.
4. If "Others" is selected under payment mode, a UI Policy dynamically displays and mandates the alternate payment field.
5. On checkout, an onSubmit client script intercepts form submission, queries the QR generation API, and renders an interactive spModal popup containing a 250x250 scannable digital entry pass.
6. Once the commuter confirms the modal, the transaction generates backend ServiceNow fulfillment records (sc_request and sc_req_item).



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Route Selection	USN-1	As a commuter on mobile, I can select my source and destination stations using reference lookups so that I can choose accurate travel stops.	Stations populate from the u_metro_station_details table without manual text input.	High	Sprint-1
Customer (Mobile user)	Fare Automation	USN-2	As a commuter, I want my journey fare calculated automatically when I pick stations and passenger counts so that I see exact ticket prices.	Fare updates immediately via an onChange client script based on Single (1) or Return (2) journeys.	High	Sprint-1
Customer (Mobile user)	Digital Ticketing	USN-3	As a commuter, I want an instant QR ticket rendered in an on-screen modal upon checkout so that I can proceed directly to gate entry.	An spModal popup appears rendering a scannable 250x250 QR code encoded with a unique ticket identifier.	High	Sprint-1
Customer (Mobile user)	Flexible Payment	USN-4	As a commuter selecting "Others" as my payment mode, I want an additional input field	UI Policy displays enter_payment_mode and enforces it as mandatory when	Medium	Sprint-1

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			so that I can specify my alternate payment provider.	mode_of_payment is set to "Others".		
Customer (Web user)	Portal Booking	USN-5	As a web user on desktop, I can book tickets through the Service Portal (/sp) catalog so that I can plan trips in advance from my computer.	Booking item renders responsively on the desktop portal with full variable support and checkout confirmation.	High	Sprint-1
Customer (Web user)	Return Journey Support	USN-6	As a web user, I can choose a return journey so that my ticket covers round trips without booking twice.	System automatically calculates and populates the return trip fare within amount_including_return.	High	Sprint-1
Customer Care Executive	Order Tracking	USN-7	As a support executive, I can track passenger booking requests via REQ and RITM records so that I can resolve commuter queries.	Generated sc_req_item records display station variables, ticket amounts, and journey types accurately.	Medium	Sprint-2
Administrator	Station Master CRUD	USN-8	As an admin, I can create, update, and manage stations in u_metro_station_details so that the station registry remains up to date.	Station admin role can manage records; updates reflect immediately in catalog station lookups.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Data Security & ACLs	USN-9	As an admin, I want role-based ACLs configured so that public portal commuters can read stations but cannot alter station database tables.	Table-level ACLs enforce read permissions for end users while restricting write/delete actions to metro_admin.	High	Sprint-1
Administrator	Fare Matrix Maintenance	USN-10	As an admin, I want to modify station fare pricing rules in the script so that ticket rates can be updated when metro policies change.	Fare conditions in client scripts can be adjusted with updated pricing rules without disrupting portal services.	Medium	Sprint-2